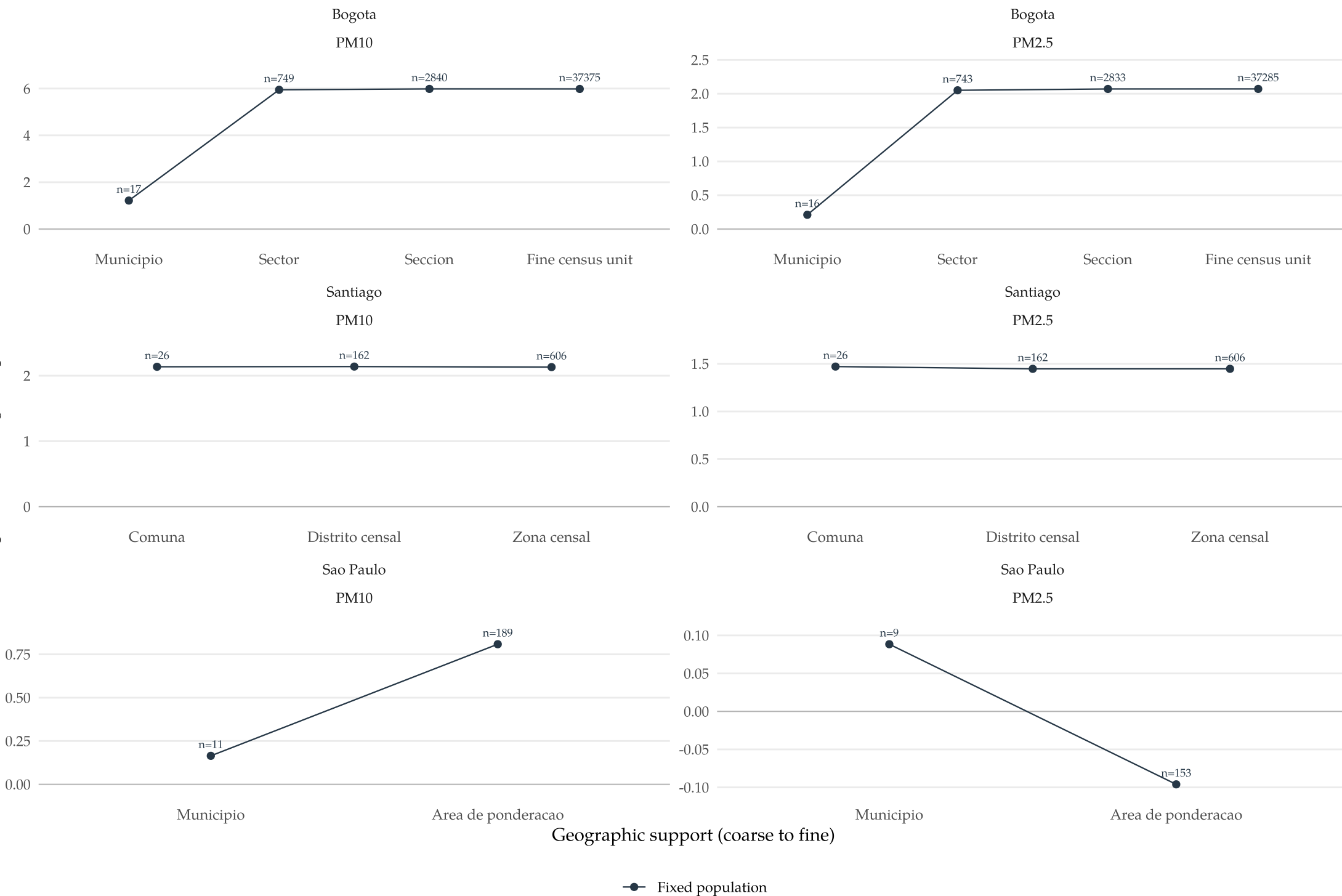


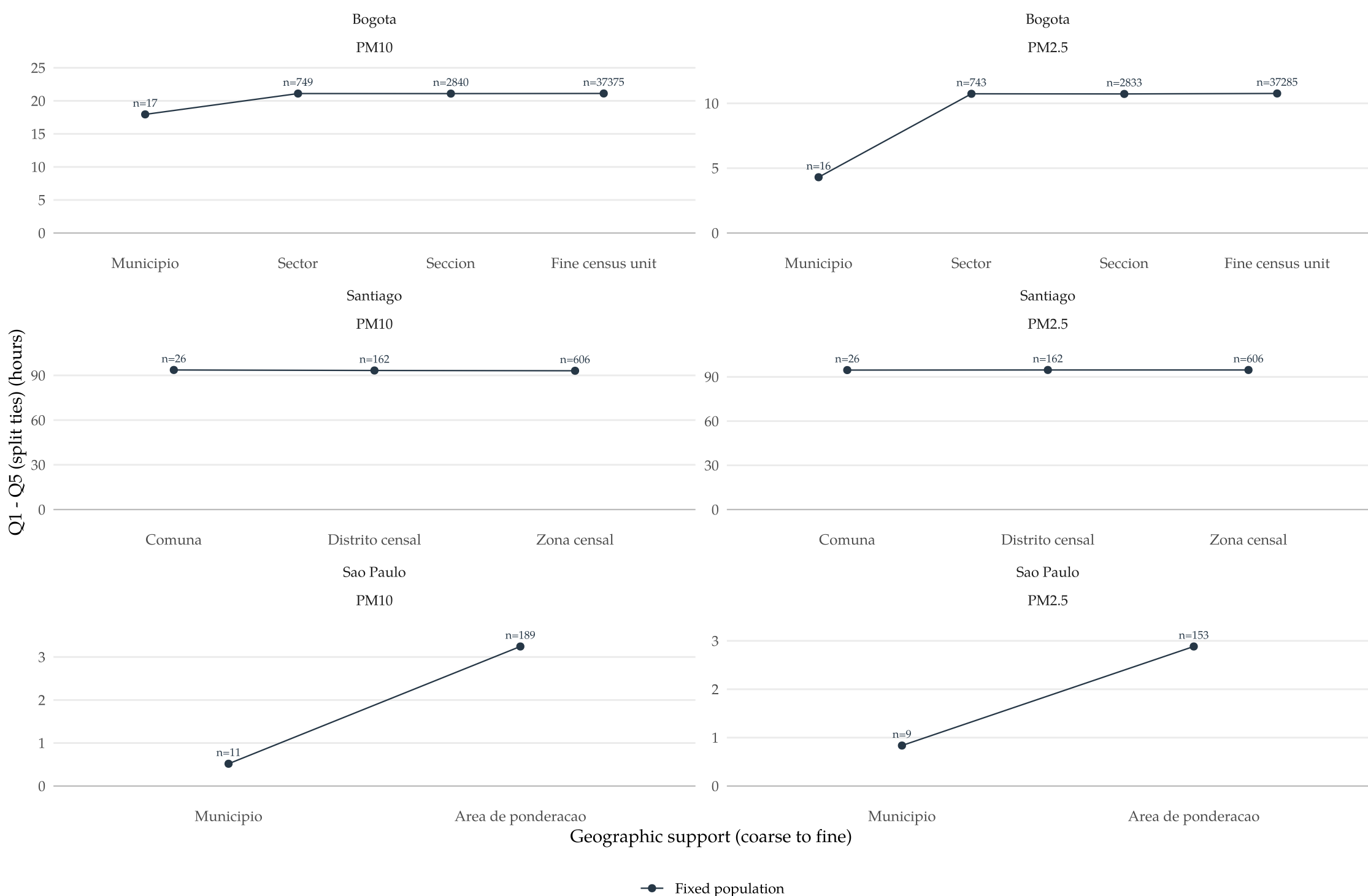
Design A - annual mean exposure

Q1 - Q5 (split ties) (micrograms per cubic metre)



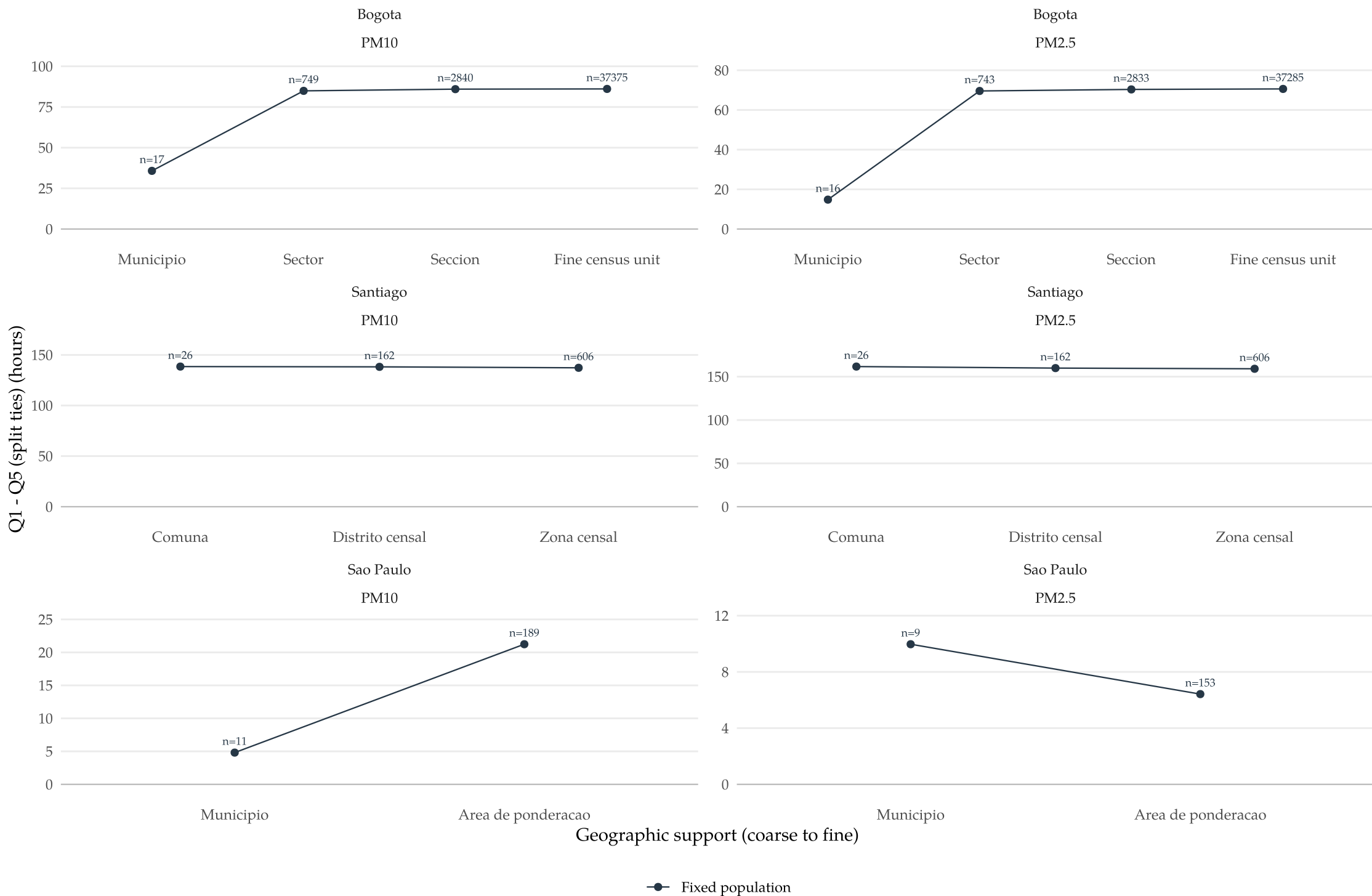
Fixed education-reporting adults and individual quintiles. Exposure aggregated with all-adult weights; locality branch shown separately.

Design A - annual hours at or above IT1



Fixed education-reporting adults and individual quintiles. Exposure aggregated with all-adult weights; locality branch shown separately.

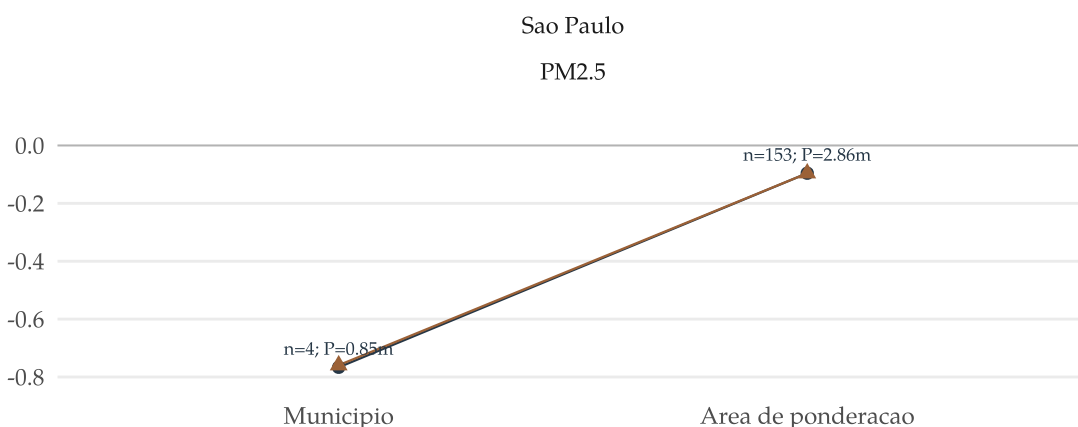
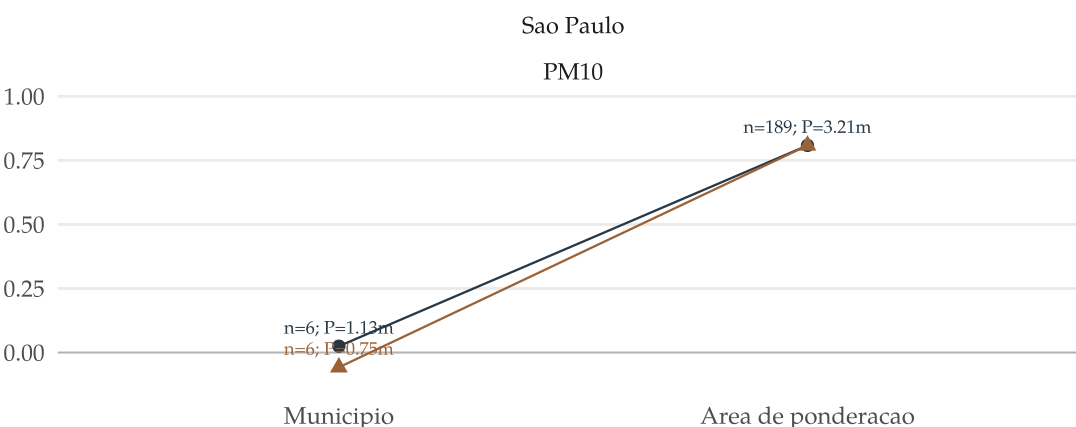
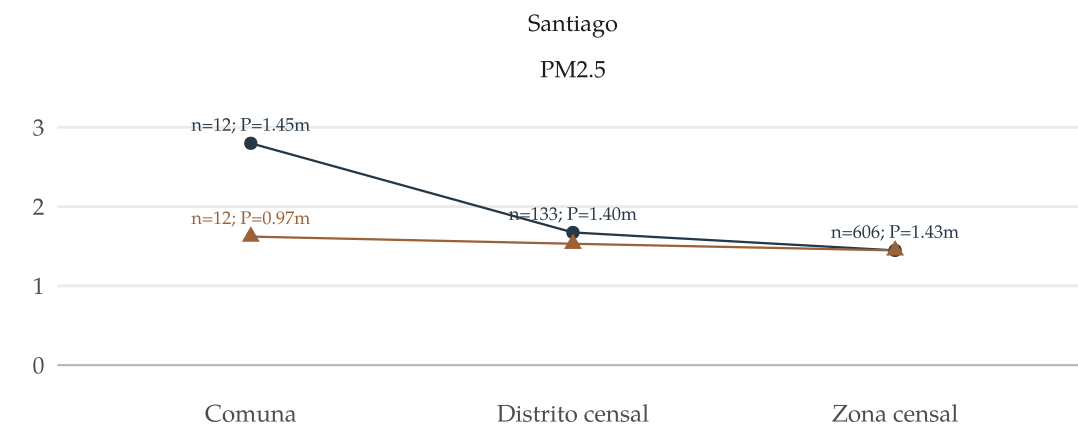
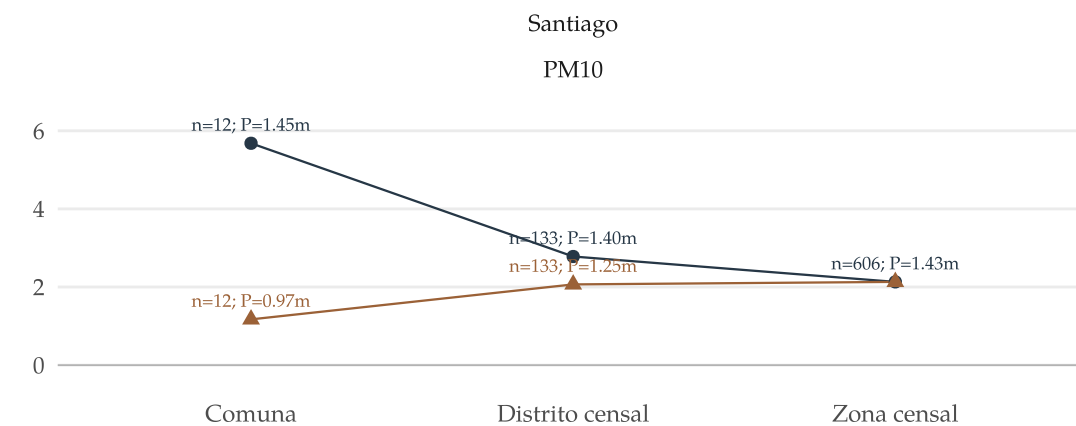
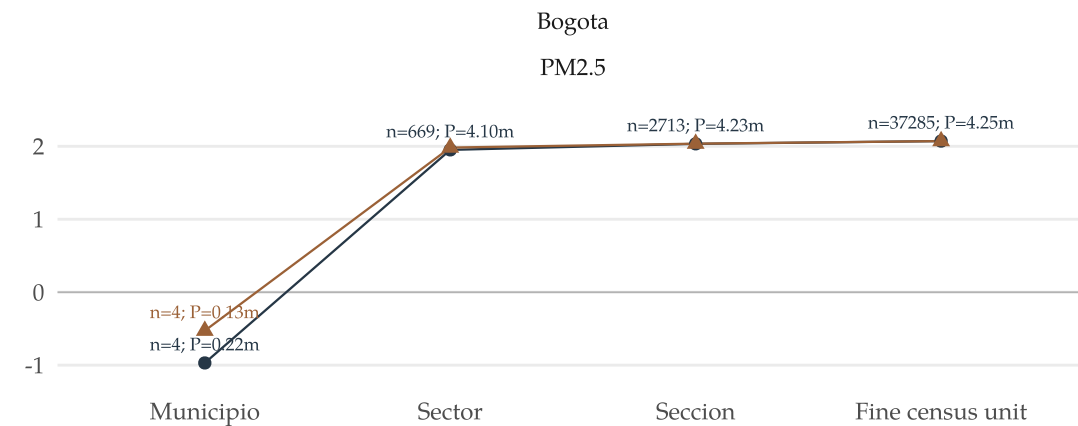
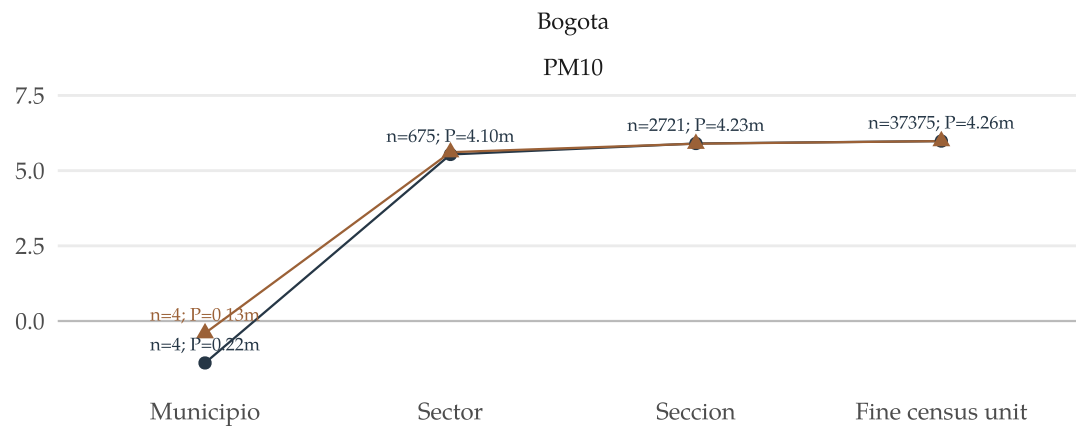
Design A - annual hours at or above IT2



Fixed education-reporting adults and individual quintiles. Exposure aggregated with all-adult weights; locality branch shown separately.

Design B - annual mean exposure

Q1 - Q5 (split ties) (micrograms per cubic metre)



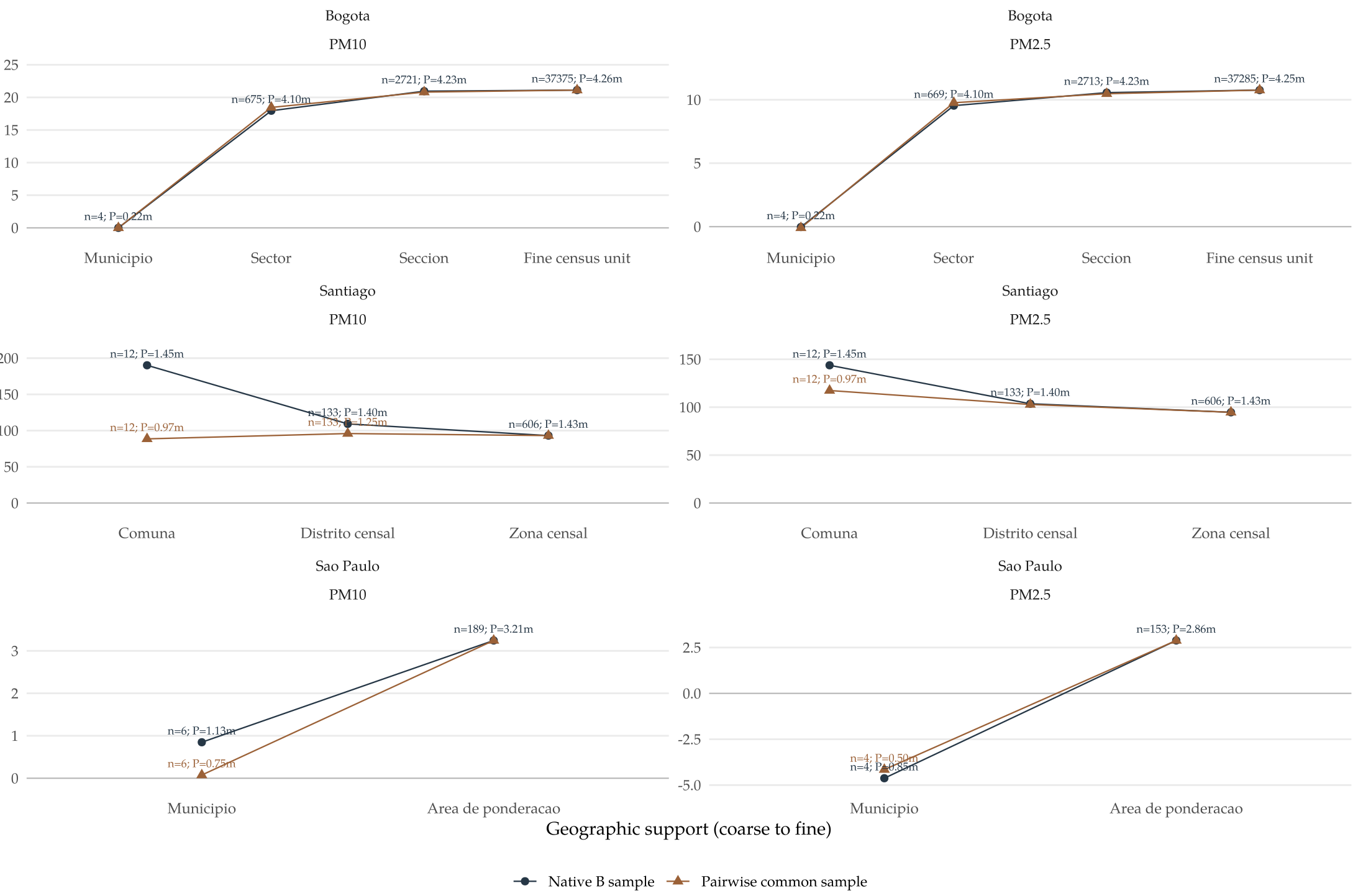
Geographic support (coarse to fine)

Native B sample Pairwise common sample

Native and common samples vary by level; P = education-reporting adults.
Common samples are pairwise intersections. Paired changes are shown separately.

Design B - annual hours at or above IT1

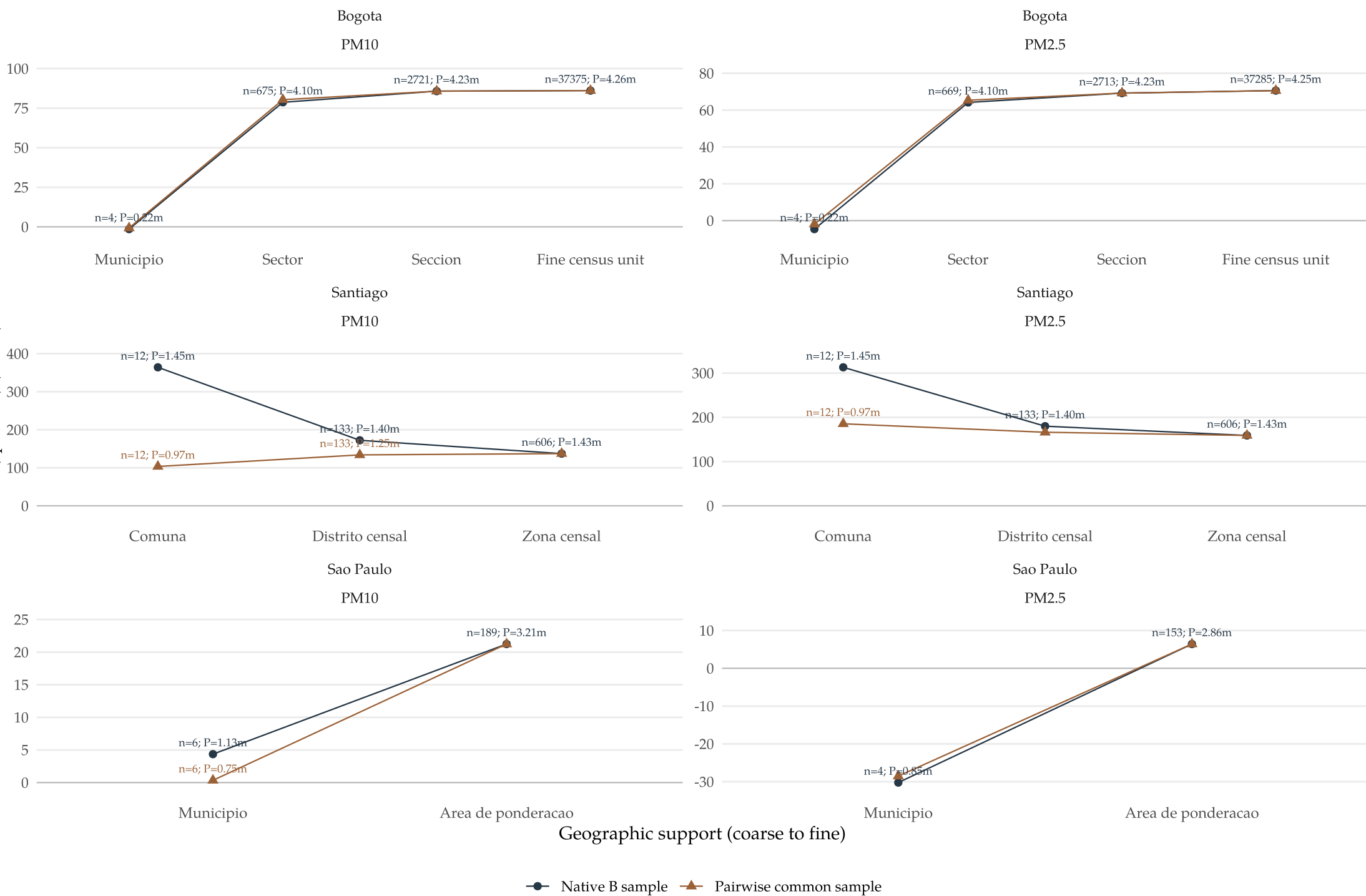
Q1 - Q5 (split ties) (hours)



Native and common samples vary by level; P = education-reporting adults.
Common samples are pairwise intersections. Paired changes are shown separately.

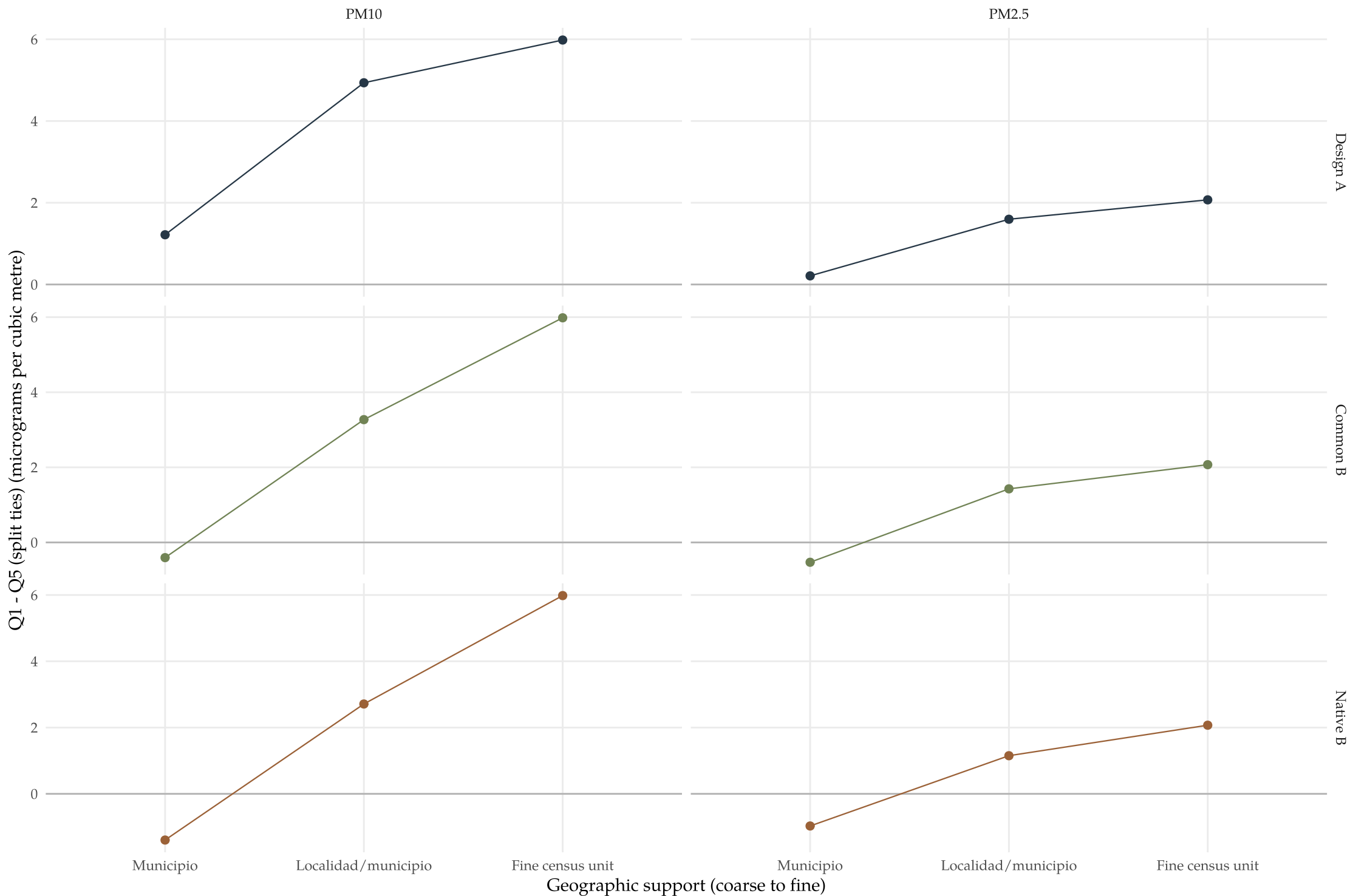
Design B - annual hours at or above IT2

Q1 - Q5 (split ties) (hours)



Native and common samples vary by level; P = education-reporting adults.
Common samples are pairwise intersections. Paired changes are shown separately.

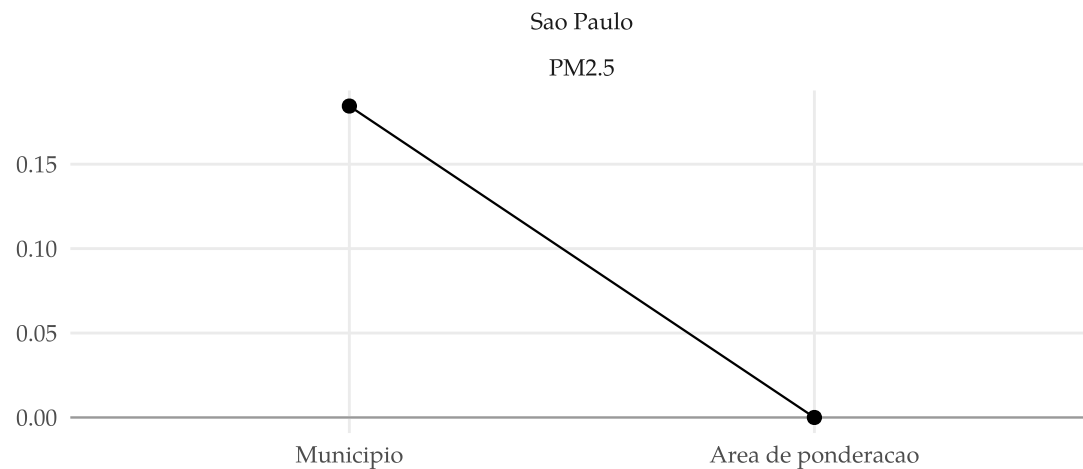
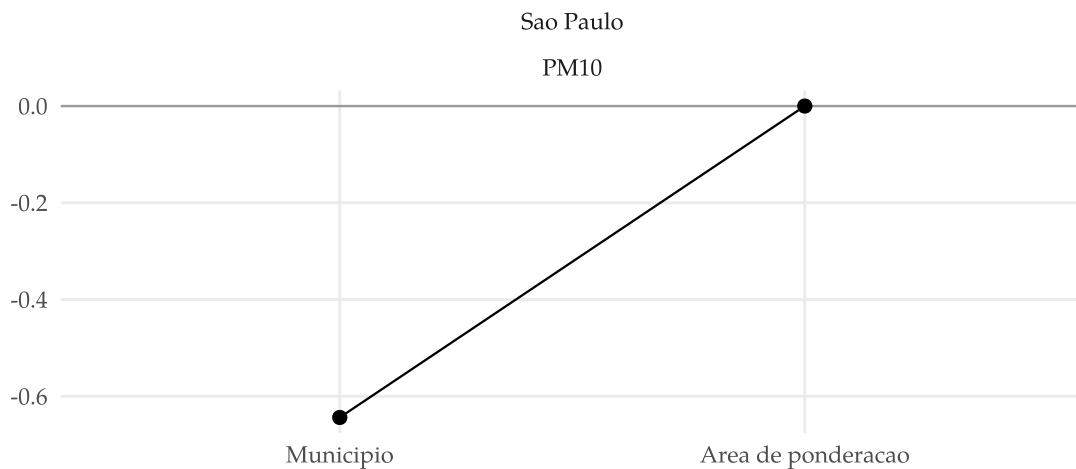
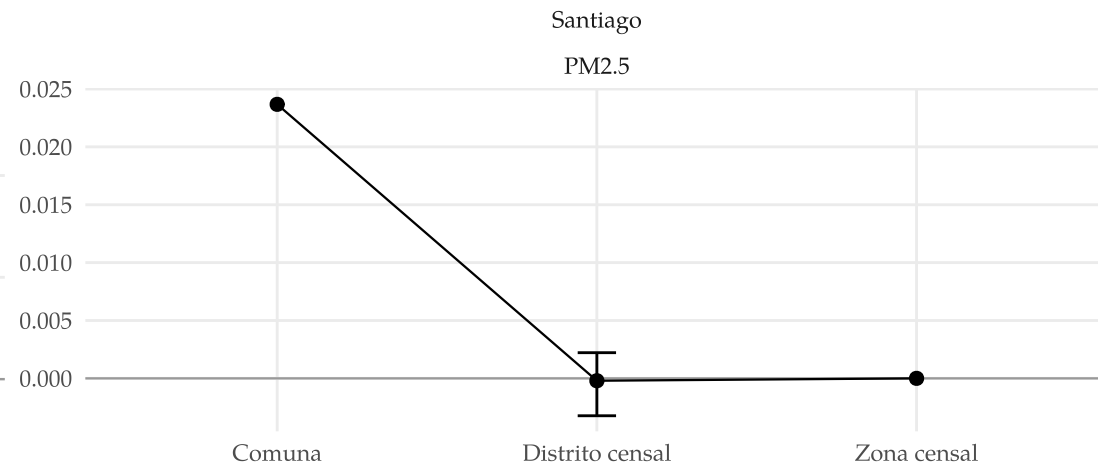
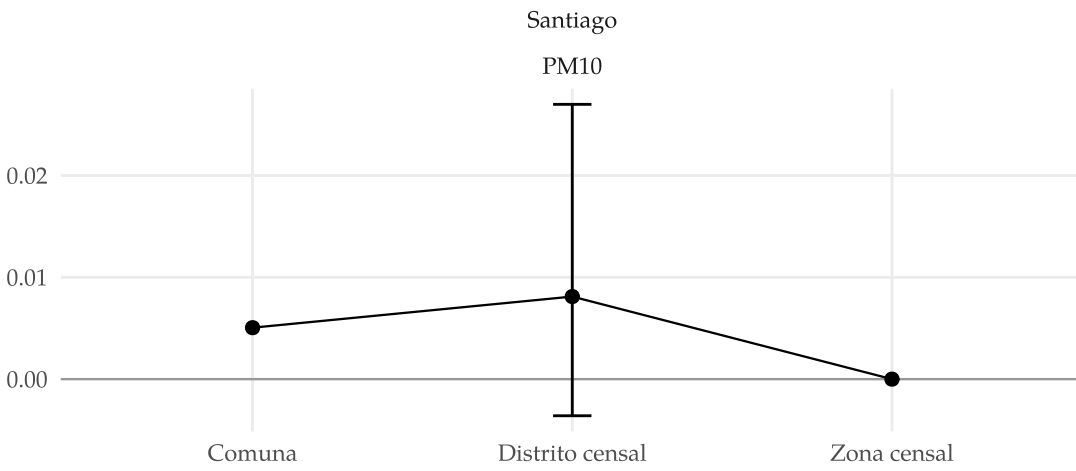
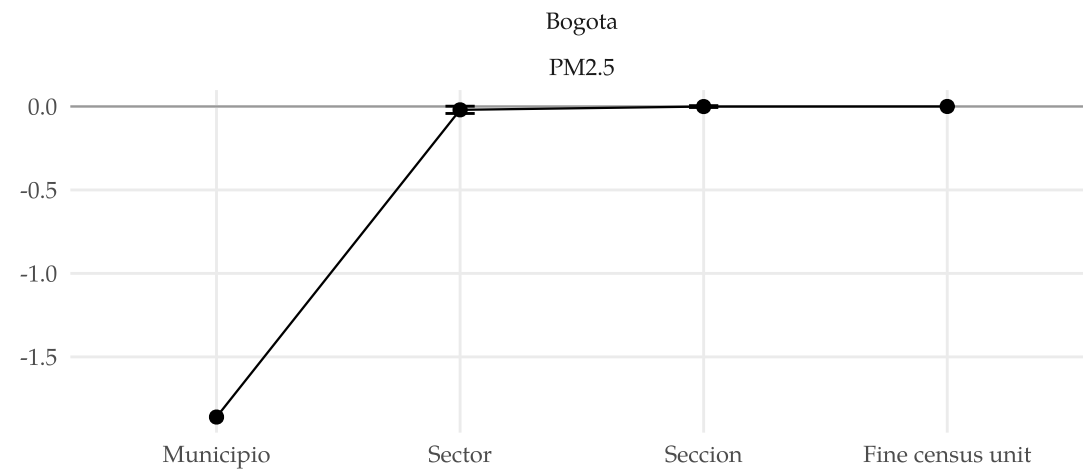
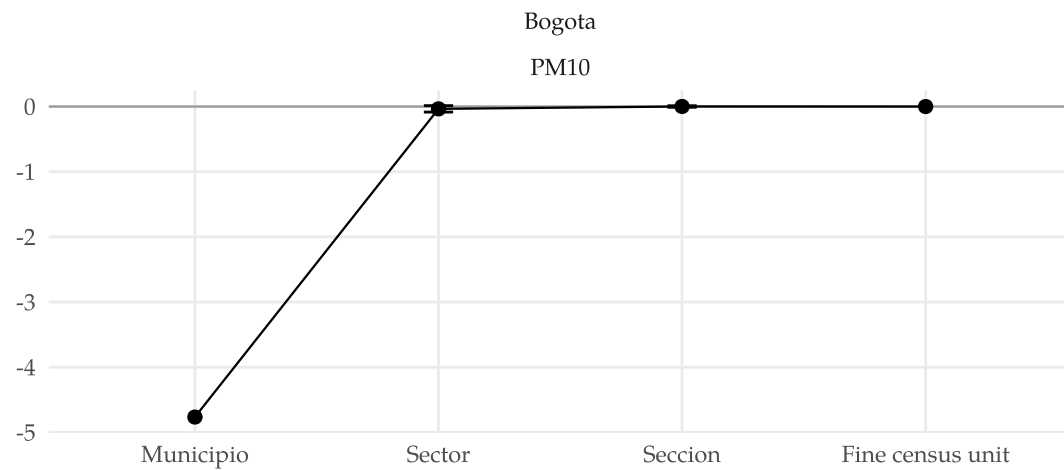
Bogota: separate locality branch



Descriptive comparisons.

Design A - change relative to finest support

Change in absolute Q1 - Q5 (split ties) gap

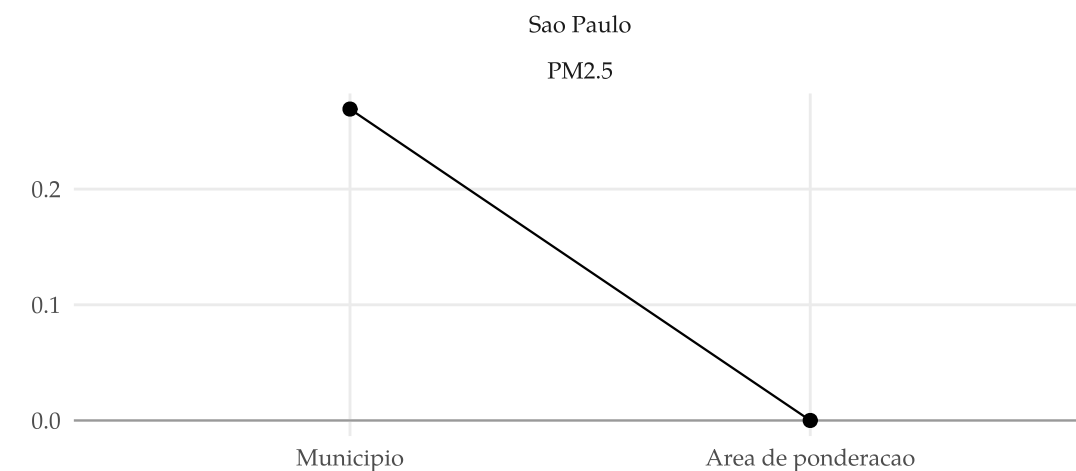
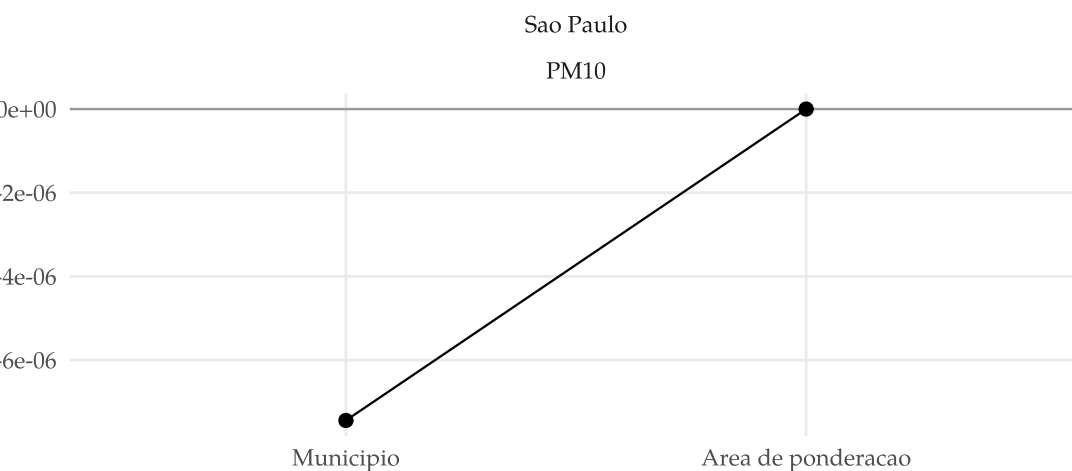
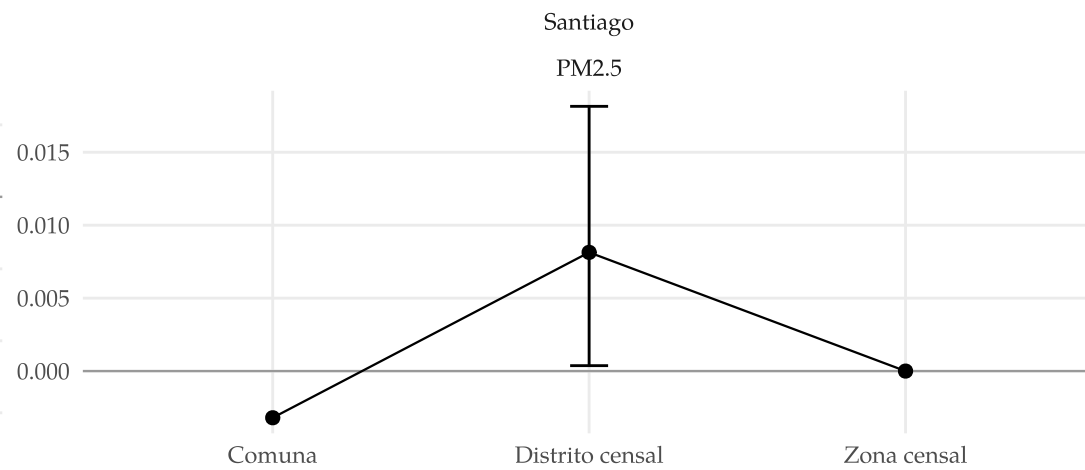
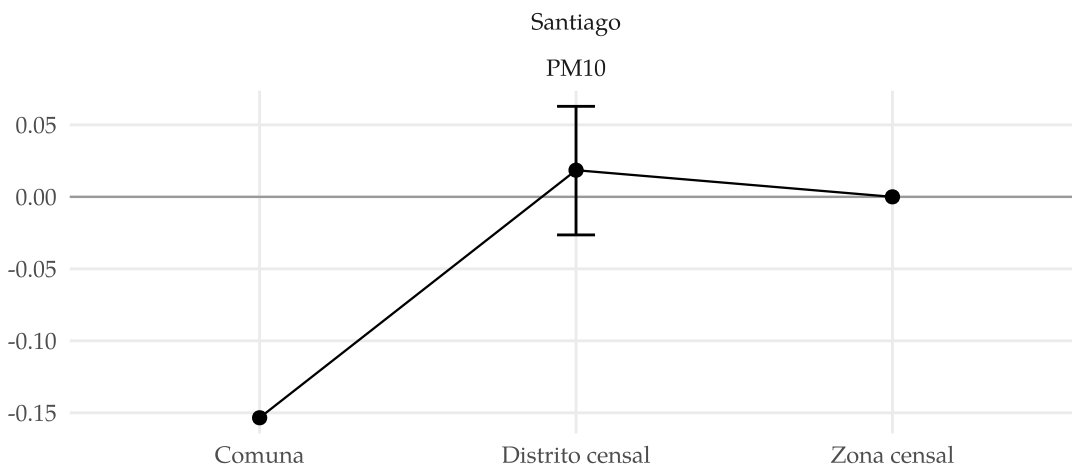
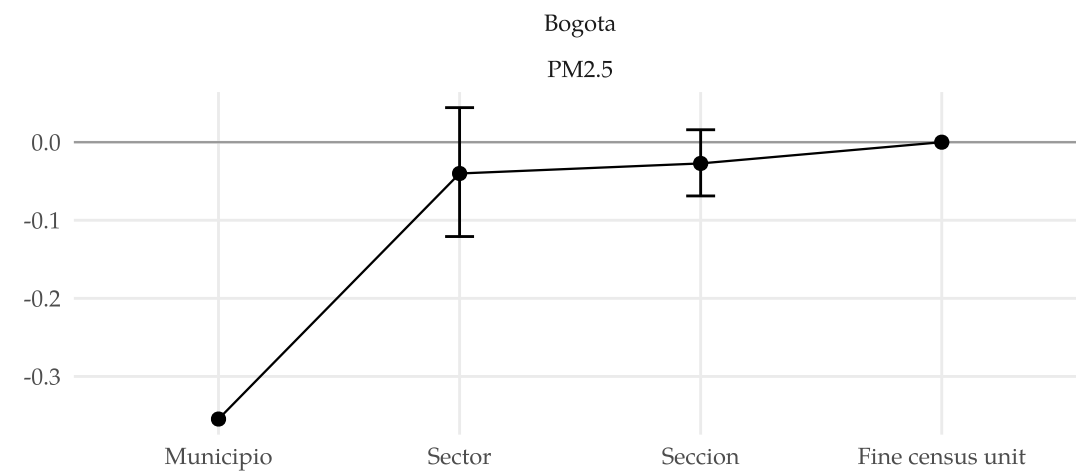
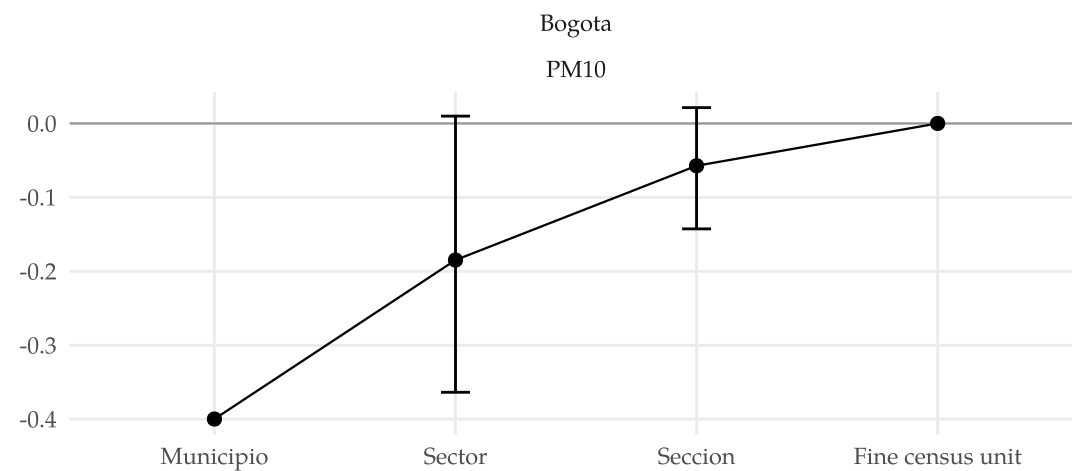


Geographic support (coarse to fine)

95% conditional geographic-cluster bootstrap intervals; 999 draws. Coarse comparisons are descriptive. B endpoints use the same pairwise population.

Common-sample B - change relative to finest support

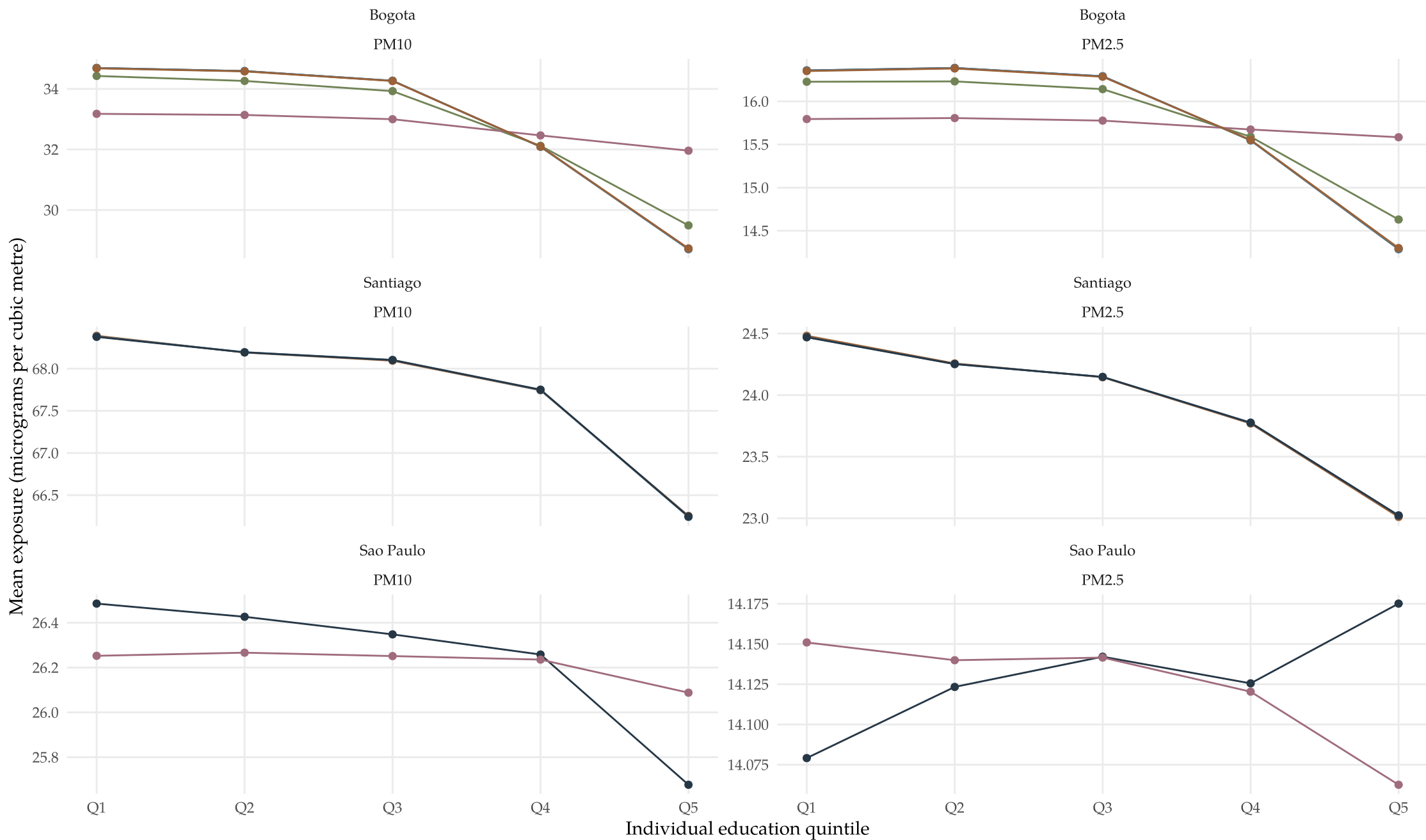
Change in absolute Q1 - Q5 (split ties) gap



Geographic support (coarse to fine)

95% conditional geographic-cluster bootstrap intervals; 999 draws. Coarse comparisons are descriptive. B endpoints use the same pairwise population.

A - full education-group profiles

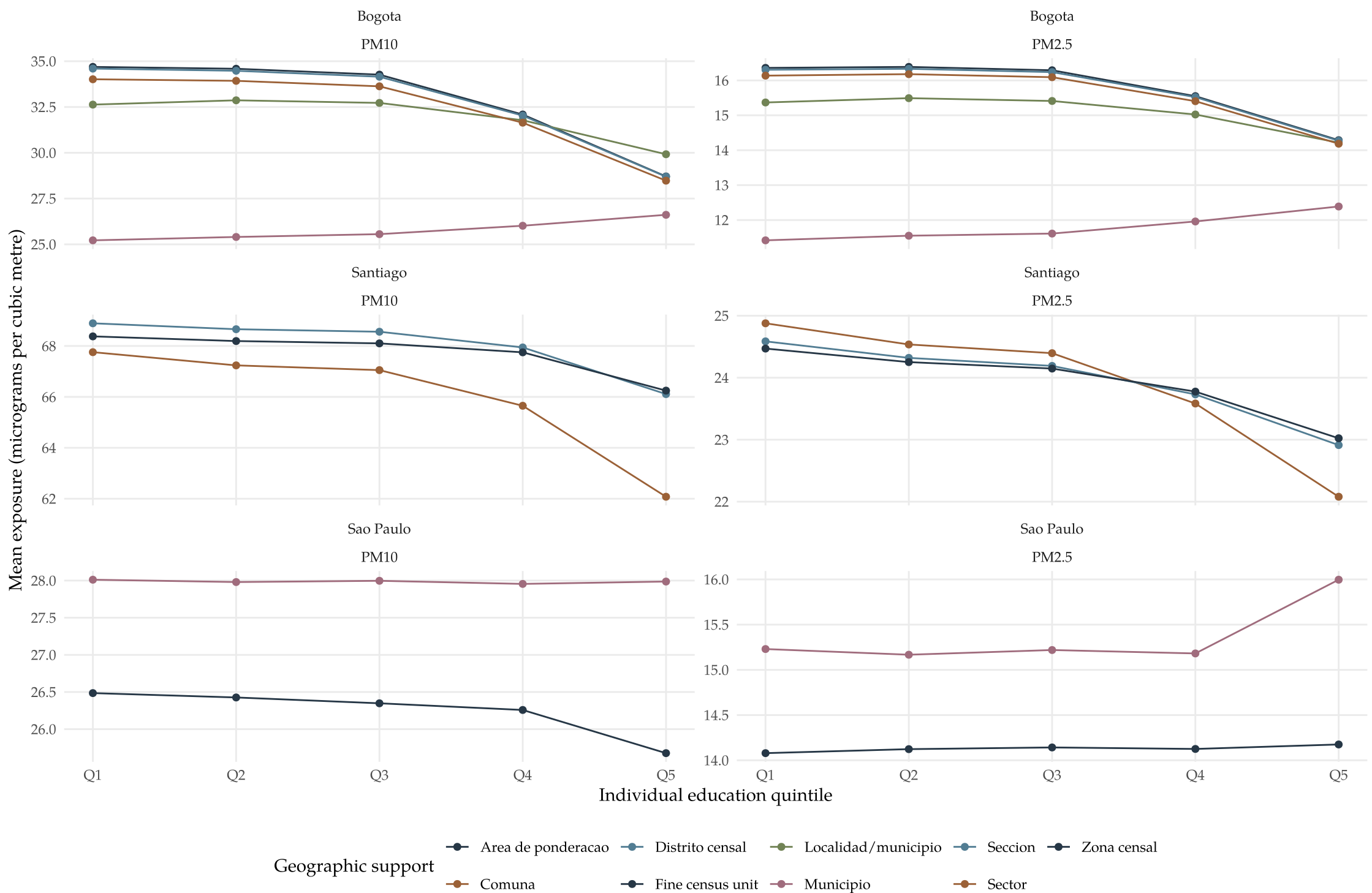


Geographic support

- Area de ponderacao
- Distrito censal
- Localidad/municipio
- Seccion
- Zona censal
- Comuna
- Fine census unit
- Municipio
- Sector

Quintiles with tied schooling values split proportionally across adjacent quintiles; group means use person weights.

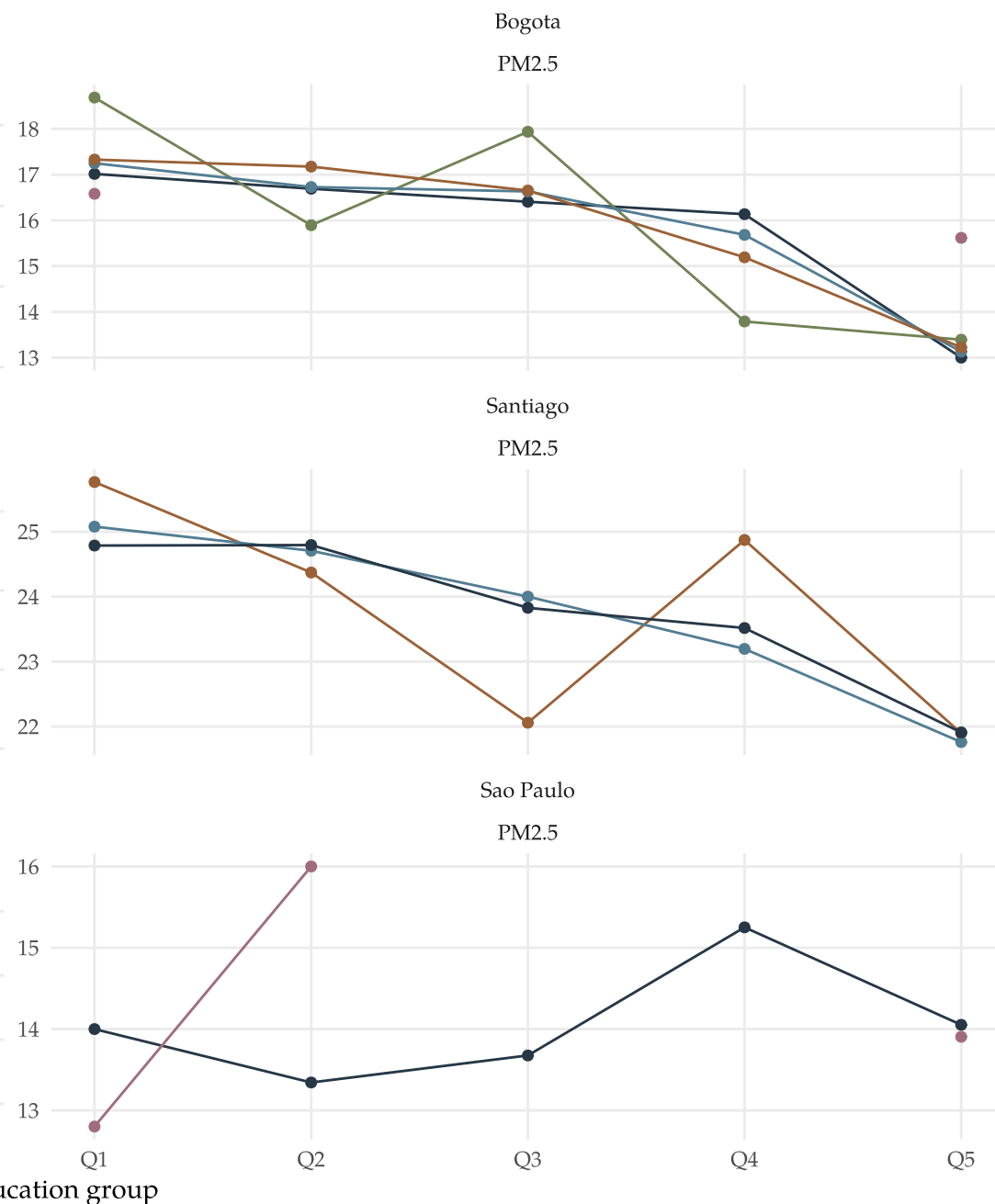
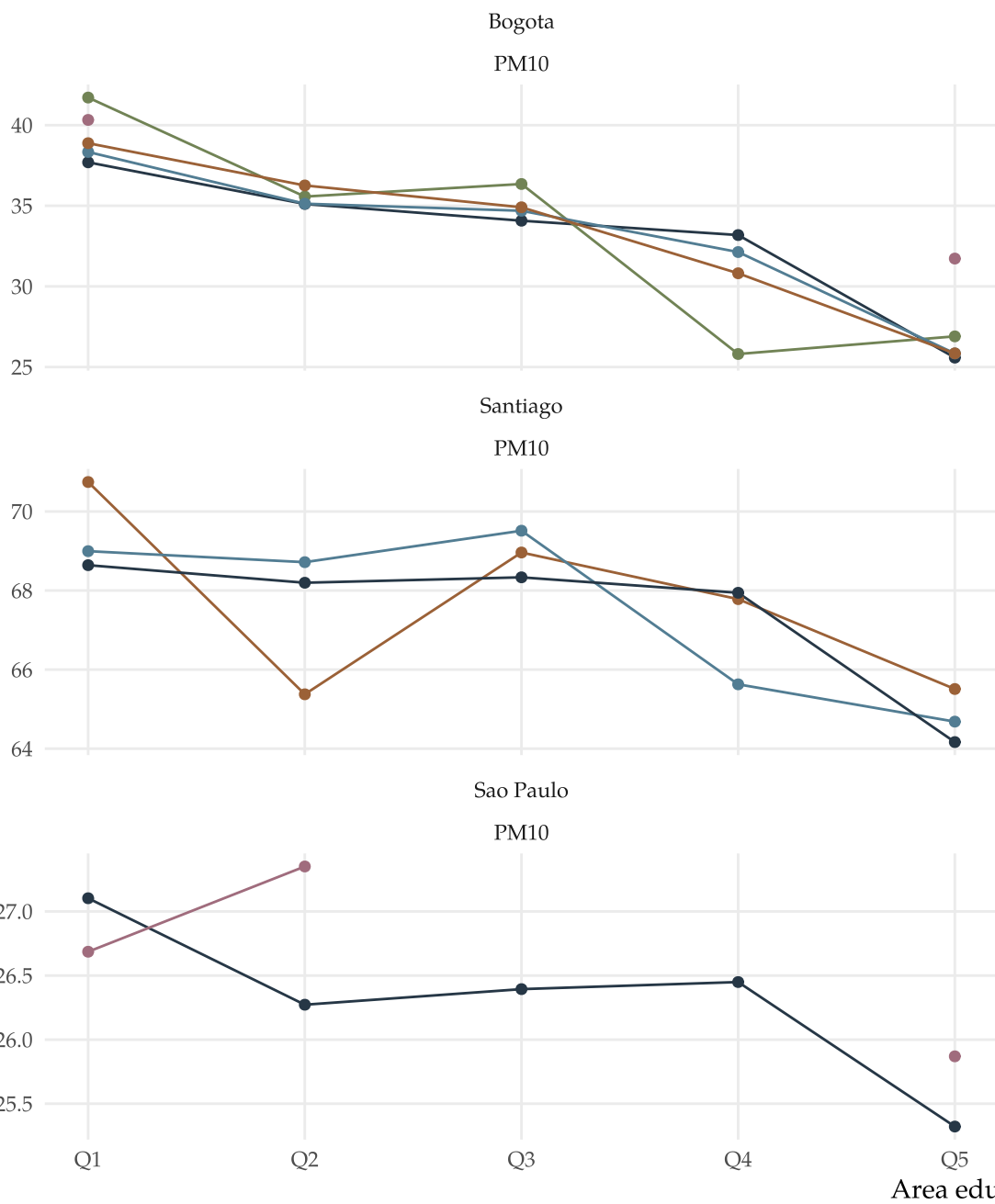
Native B - full education-group profiles



Quintiles with tied schooling values split proportionally across adjacent quintiles; group means use person weights.

C - full education-group profiles

Mean exposure (micrograms per cubic metre)

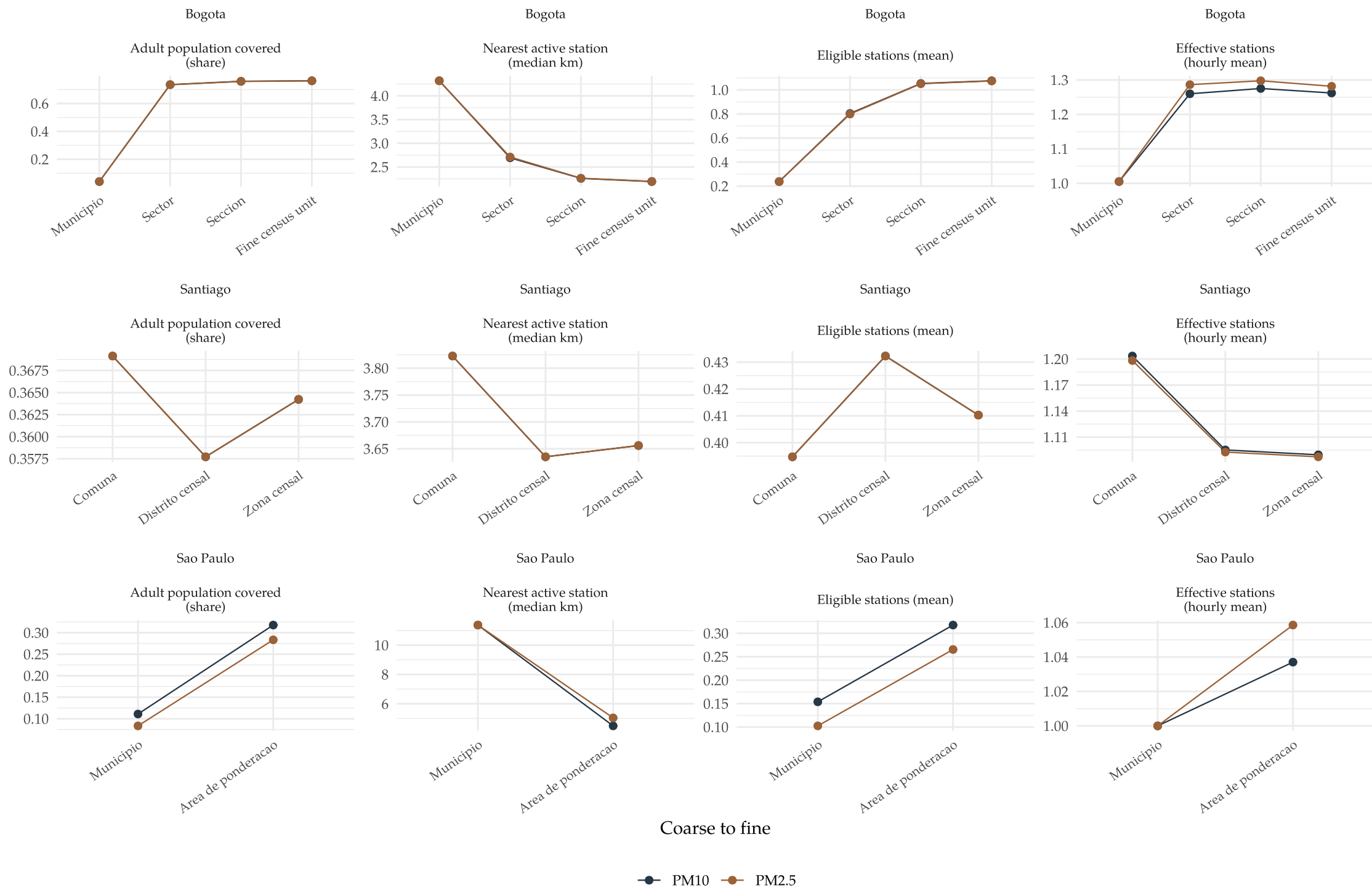


Geographic support

- Area de ponderacao
- Distrito censal
- Localidad/municipio
- Seccion
- Zona censal
- Comuna
- Fine census unit
- Municipio
- Sector

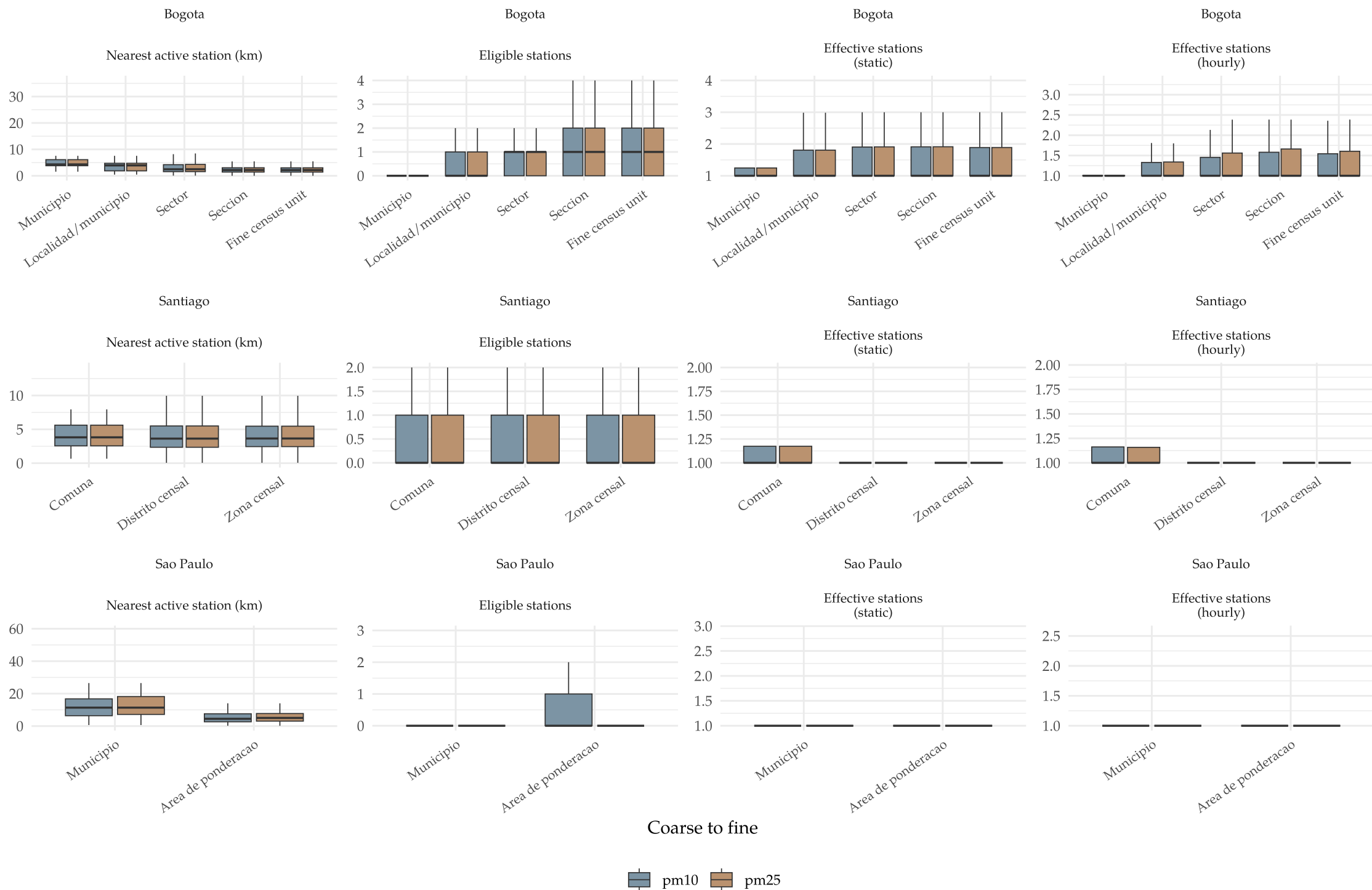
Classification only; finest exposure fixed. Empty area groups remain unavailable. These groups differ from A/B quintiles.

Monitoring-matrix sensitivity under Design B



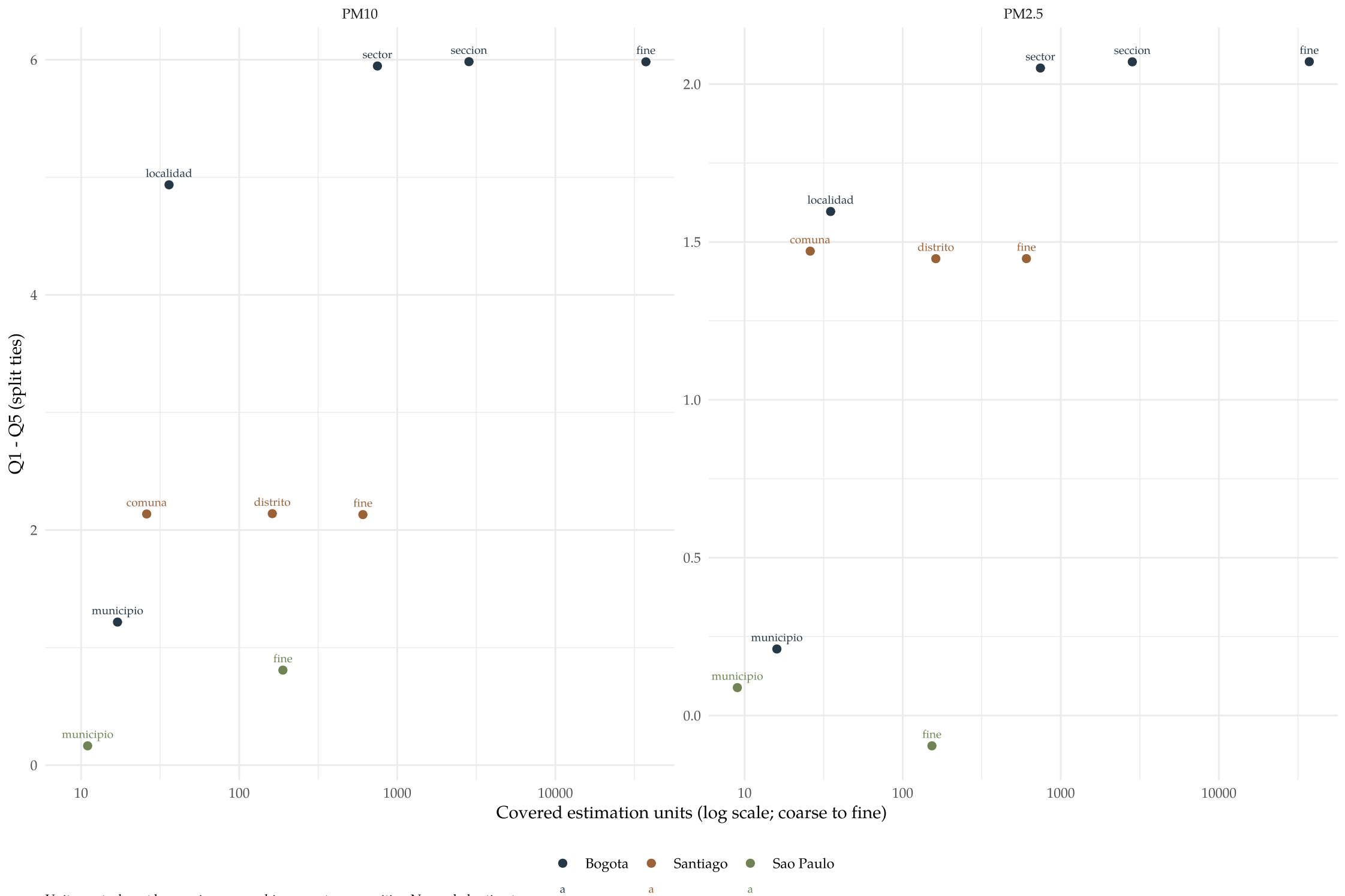
Catalog, active, eligible and contributing station counts are distinct in the tables.

Monitoring-matrix distributions across populated units



Boxes: unweighted unit quartiles; whiskers: 1.5 IQR. Outliers omitted from view only.

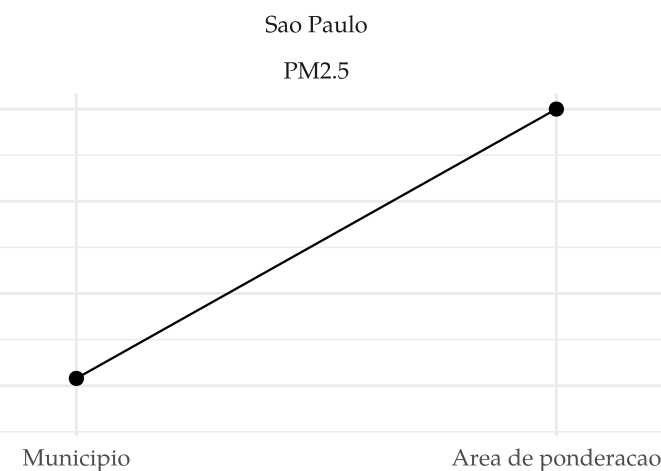
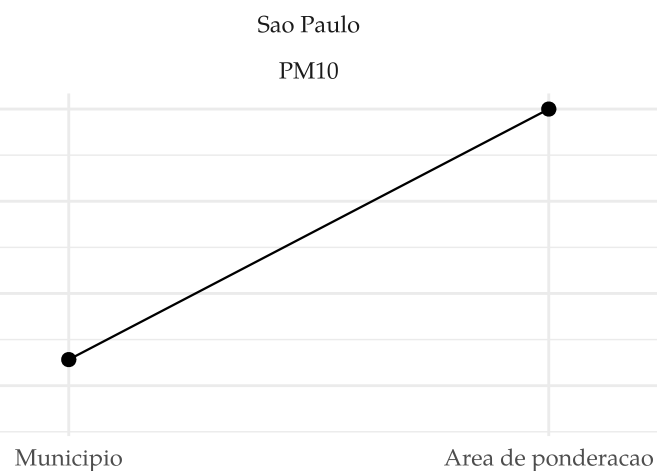
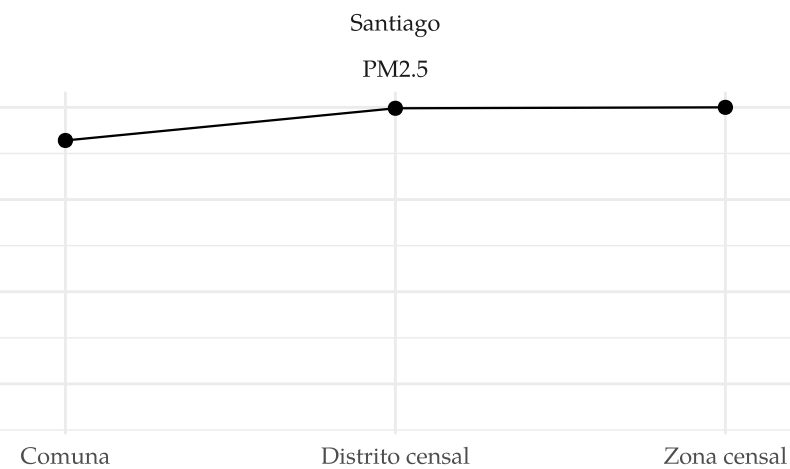
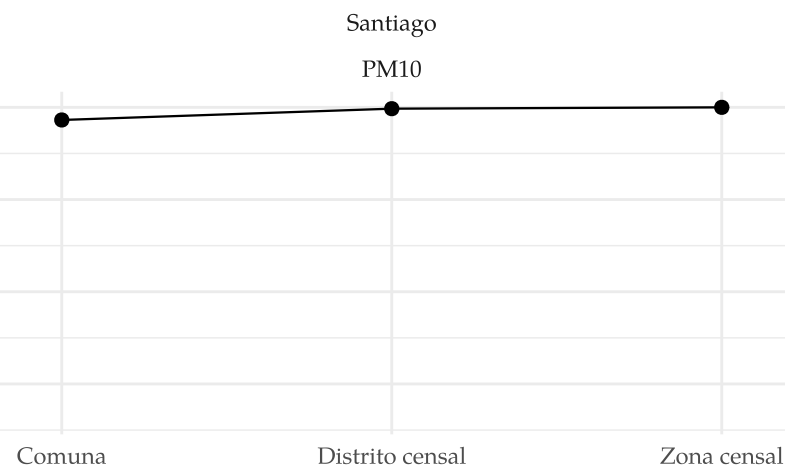
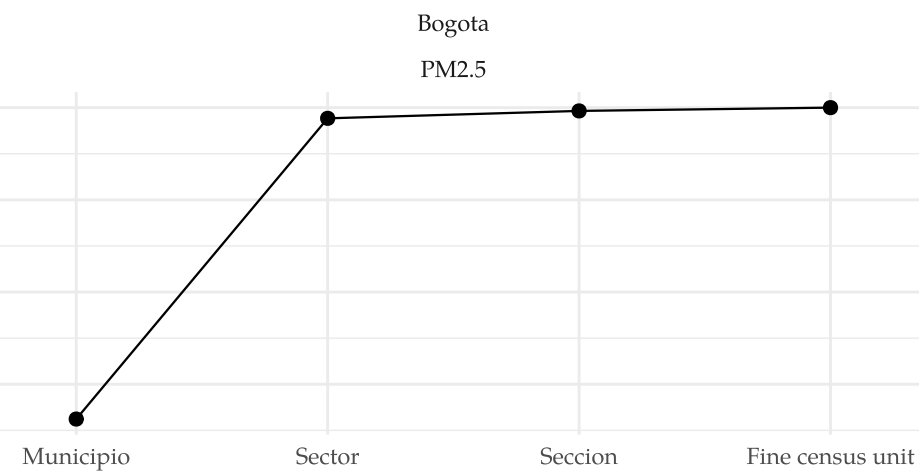
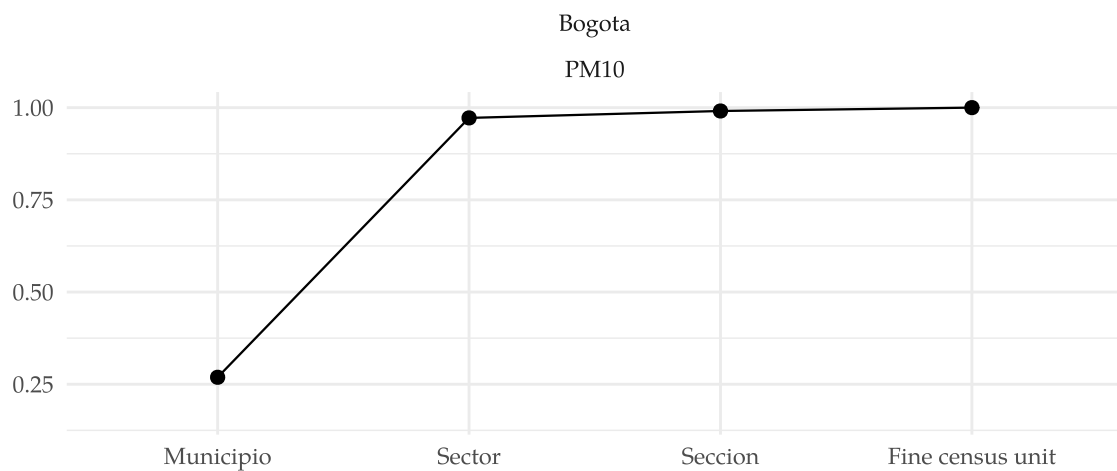
Design A: covered-unit counts as a supplementary resolution measure



Unit counts do not harmonize geographic concepts across cities. No pooled estimate.

Design A: exposure variance retained

Share of finest-level variance



Coarse to fine

All-adult aggregation weights and fixed covered support. Declining variance does not imply a monotone gap.